

ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 4TH YEAR

FINAL YEAR PROJECT

PROJECT GUIDE

MARKS: 100 (see the marking rubric) **TIME: Full academic year**

BEng Mechatronics and Robotics Engineering, 4th year: a step-by-step guide to the final-year design project, from choosing a topic and writing the proposal to building, testing, writing the report and the oral examination.

Written by Academic Central for practice, following final-year design project modules at South African universities (e.g. University of Pretoria EPR 402 / EPR 405 Project: a major individual design project from concept to delivery, with a proposal, technical report, oral examination and demonstration) and the ECSA graduate attributes for the BEng degree. It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This guide follows the way South African universities run the final-year design project. Your own university's study guide, dates and rules always come first.
2. The project is individual work. Your supervisor guides you, but every design decision, calculation and line of the report must be your own.
3. The marking rubric (the "memo" of this guide) shows how a typical project is assessed. Read it before you start.

1. WHAT THE PROJECT IS

The final-year project is usually the biggest module of the degree (at the University of Pretoria, for example, it is worth 64 credits). You take one engineering problem from an idea to a working product. You must: formulate the problem; write technical requirements and specifications in a formal proposal; plan the work; design from first principles; build and test the system; show that it meets the specifications in real conditions; and document everything in a technical report that you defend in an oral examination and a demonstration.

The project is where the Engineering Council of South Africa (ECSA) checks that you can work as an engineer. Several graduate attributes (GAs) are assessed here, and many universities require you to pass each one separately (a subminimum), so a good demonstration cannot make up for a weak report.

2. THE ECSA GRADUATE ATTRIBUTES YOU MUST SHOW

GA	Attribute	How the project shows it
GA1	Problem solving	You turn a vague need into a clear engineering problem and solve it.
GA2	Application of scientific and engineering knowledge	Your design uses real theory: dynamics, circuits, control, programming.
GA3	Engineering design	Specifications, concepts, selection, calculations, detailed design.
GA4	Investigations, experiments and data analysis	Planned tests, measurements, results and conclusions.
GA5	Engineering methods, skills and tools, including IT	CAD, simulation, microcontroller tools, test equipment.
GA6	Professional and technical communication	The proposal, the report, drawings and the oral examination.
GA7	Sustainability and impact of engineering activity	Safety, cost, environment and people affected by your design.

GA8	Individual, team and multidisciplinary work	Working with your supervisor, workshop staff and others.
GA9	Independent learning ability	Learning new topics and tools on your own.
GA10	Engineering professionalism	Honesty, ethics, referencing, meeting deadlines.
GA11	Engineering management	Planning, budget, risk and time management.

Check your study guide: each university says which GAs this module assesses. Design (GA3), investigation (GA4) and communication (GA6) are almost always included.

3. A TYPICAL TIMELINE

When	What you do	What you hand in
Weeks 1 to 3	Choose a topic and a supervisor; read about the problem.	Topic form
Weeks 4 to 6	Write requirements and specifications; plan the work and the budget.	Project proposal
Weeks 7 to 12	Literature study; generate and select concepts; first calculations and simulations.	Concept or progress report
Mid-year	Present your design; order parts early.	Mid-year presentation
Third term	Detailed design, build, program and test each part (module testing).	Lab book, progress meetings
Early fourth term	Integrate the whole system; run the final tests against every specification.	Test results
Fourth term	Write the report; prepare the demonstration.	Final report
End of year	Demonstrate the working system and defend your work.	Oral examination and demo

Parts can take weeks to arrive, and something always breaks. Plan at least three weeks of spare time before the report is due.

4. CHOOSING A GOOD MECHATRONICS PROJECT

A good project is: big enough to need real engineering design from first principles; small enough to finish in one year; measurable (you can test whether it works); affordable within your budget; and something you find interesting. It should combine mechanics, electronics and software, which is what mechatronics is.

- A self-balancing robot with a microcontroller, IMU and a designed control loop.
- A robot arm that sorts objects by colour with a camera (machine vision) on a conveyor.
- An automated greenhouse or irrigation controller that saves water, with sensors and a mobile app.
- A line-following or mapping delivery robot for a warehouse or hospital.
- A low-cost prosthetic hand controlled by muscle (EMG) signals.
- A solar tracker or battery-monitoring system that keeps working during load shedding.
- A vision system that inspects products (for example bottle caps or fruit) and rejects bad ones.

Avoid projects where you only connect ready-made modules and copy example code. The examiners look for YOUR design: your calculations, your circuit, your control loop, your mechanical parts.

5. THE PROJECT PROPOSAL

The proposal is a short formal document (often 5 to 10 pages) that your supervisor approves. Use these headings:

- **Problem statement:** the need, who has it, and why it matters.
- **Requirements:** what the system must do, in words.
- **Specifications:** measurable values, each with a test method (see the example below).
- **Scope:** what you will and will not do; what you design yourself and what you buy.
- **Approach and concepts:** how you plan to solve it, with first ideas.
- **Plan:** a Gantt chart with milestones.
- **Budget:** a parts list with prices and suppliers.
- **Risks:** what could go wrong and your back-up plan.
- **References.**

Example of good specifications for a self-balancing robot:

No.	Specification	Value	How it will be tested
S1	Stays upright on a flat floor	at least 10 minutes	Timed test, 5 runs
S2	Recovers from a push	tilt of up to 10°	Push test with the angle logged from the IMU
S3	Top speed	at least 0.5 m/s	Time over a 5 m track
S4	Battery life	at least 1 hour	Run until the low-battery cut-off
S5	Total cost of parts	at most R2 500	Final parts list

Words like "fast", "accurate" or "user-friendly" are not specifications. If you cannot test it, rewrite it.

6. DESIGN, BUILD AND TEST

- Generate at least two or three concepts for each main part and choose with a weighted decision matrix.
- Design from first principles: show the calculations for motor sizing, gear ratios, power budget, sensor resolution, controller gains, strength of parts.
- Model and simulate before building (for example MATLAB/Simulink, Python, SPICE or CAD).
- Keep a lab book (paper or digital) with dates, decisions, measurements and problems.
- Test each module on its own before joining them (module tests), then test the whole system (integration).
- Plan the final tests from your specification table, so every specification has a result.
- Use version control (for example Git) for your code and back up your work every day.

7. THE FINAL REPORT

A typical final report is 40 to 60 pages plus appendices. Follow your department's template and page limit. A common structure:

Part	What it contains
Summary	One page: the problem, what you built, and the main results against the specifications.
1. Introduction	Background, problem statement, objectives, scope and the layout of the report.
2. Literature study	Existing solutions and the theory you need, properly referenced.
3. Requirements and specifications	The final specification table, with any changes from the proposal explained.
4. Design	Concepts and selection, then the detailed mechanical, electronic and software design with calculations.
5. Implementation	How it was built and programmed; photographs, drawings, circuit diagrams, flowcharts.
6. Results	Test method, measurements and results for every specification, with graphs and tables.
7. Discussion	What worked, what did not and why, impact, safety and cost.
8. Conclusion and recommendations	Were the objectives met? What should be improved next?
References and appendices	References in one style; code, drawings, datasheet extracts, the parts list.

8. THE ORAL EXAMINATION AND DEMONSTRATION

- Prepare a short presentation (often 10 to 15 minutes): problem, specifications, design, results.
- Show the system working. Have a back-up video in case something fails on the day.
- Expect questions on every calculation and every choice: "Why this motor?", "How did you choose these gains?", "What would you change?"
- Be honest about what does not work and explain why. Examiners respect a clear explanation more than excuses.

9. COMMON MISTAKES TO AVOID

- Choosing a project that is too big, or starting to build before the design is done.
- Specifications that cannot be measured, or tests that do not match them.
- Ordering parts too late, and having no plan B.
- Copying code or text without referencing it (plagiarism is a serious offence).
- Leaving the report for the last two weeks.
- Showing only that it works, without measurements to prove it.

10. CHECKLIST BEFORE YOU HAND IN

- Every specification has a measured result in the report.
- Every design choice is backed by a calculation, a simulation or a test.
- All figures and tables are numbered, labelled and referred to in the text.
- All sources are referenced; your code and drawings are in the appendices.
- The report follows the template and the page limit, and has been proofread.
- Your supervisor has seen a draft in time to comment.